PROBLEM 33

Two Repelling Particles on a Rotating Parabola

Original statement · Problems.pdf, p. 21

Rotating Wire + Two Particles

Two heavy point particles, P_1 and P_2, of mass m each, slide without friction along a rigid wire parabola given by the equation $z = x^2/(2L)$, where L is a known length. The wire rotates about its axis with a fixed, known angular velocity ω. The two particles repel each other electrically with a force of magnitude $F_E = k/d^2$, where k is a known constant and d is the distance between the particles. We define the rotating reference frame $Oxyz$ attached to the parabola as shown in the figure below.

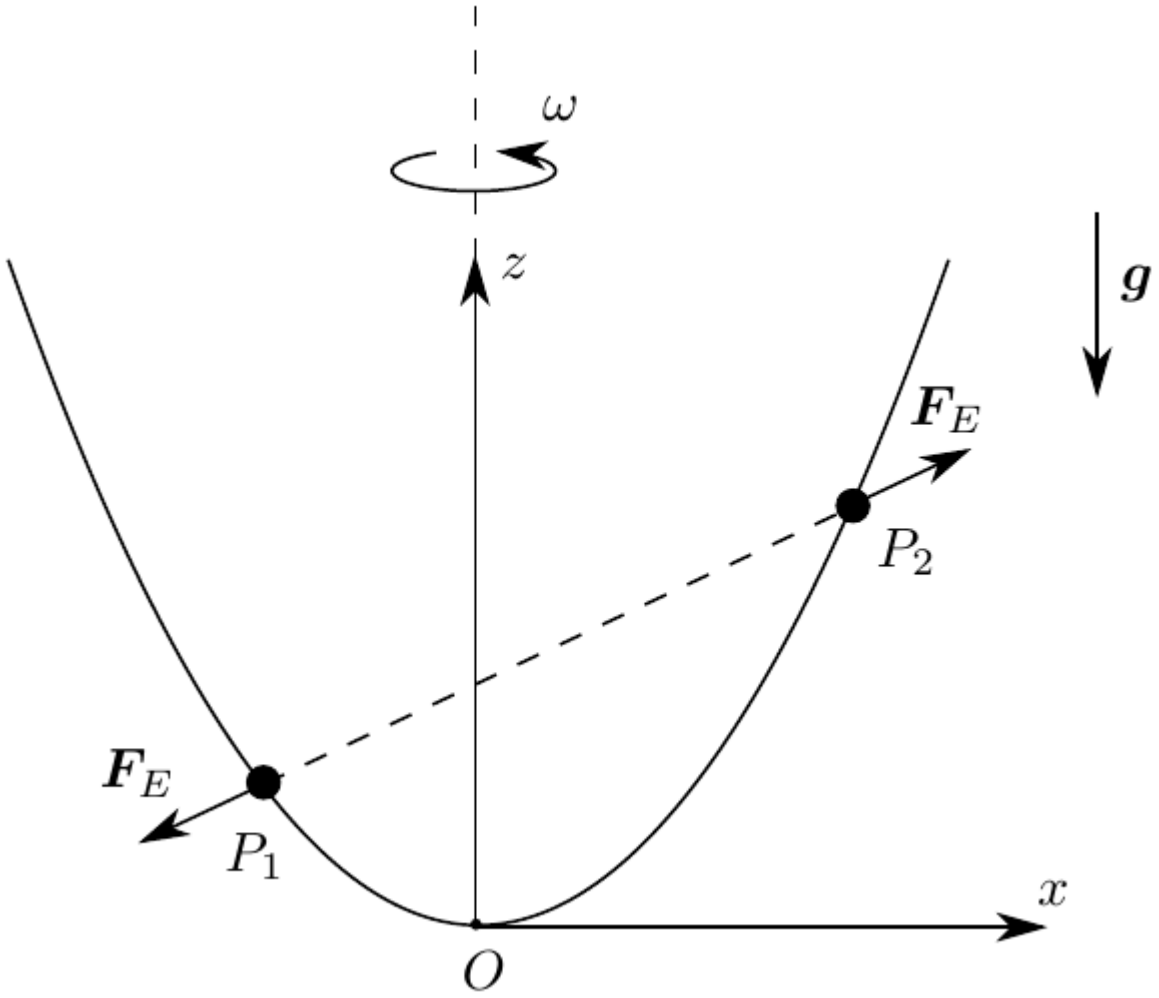


Figure: Parabola z = x²/(2L) in the plane Oxz rotating with angular velocity ω about Oz, particles P1 (left branch) and P2 (right branch) with the repulsive forces F_E along the line P1P2, gravity g downward (Problems.pdf, p. 21).

(a) Write down the position, velocity, and acceleration vectors of each particle relative to $Oxyz$ in a generic instant of time, as a function of x_1, x_2 (the position of P_1 and P_2 along the Ox axis) and their derivatives.

(b) Write down Newton's second law for each particle in the $Oxyz$ reference frame.

(c) If the rotation of the wire is maintained by a motor, how much torque must it produce in the Oz direction to keep ω constant as a function of the variables of the problem and their derivatives?

(d) Find out the equilibrium configurations of the system in the reference frame $Oxyz$ as a function of the parameters $\gamma = g/(\omega^2 L)$ and $\kappa = k/(m\omega^2 L^3)$.

Consider now that P_1 and P_2 are constrained to move symmetrically on each branch of the parabola, and that $\gamma = \kappa = 2$. Initially, $x_1(0) = L$ and $v_1(0) = L\omega$ (with $\dot{x}_1(0) > 0$).

(e) Write down the equation that gives the minimum and maximum values of x_1 occurring in this case. *Hint: this equation is a third-order polynomial in x_1.*

Worked Solution

Reference Frames, Bases and Coordinates

- **System:** two particles P_1, P_2 of mass m slide on a smooth parabolic wire $z = x^2/(2L)$ that spins about its vertical axis Oz at constant ω. They repel each other with $F_E = k/d^2$ along the line that joins them, and both are heavy.
- **Why the rotating frame $Oxyz$.** In $Oxyz$ the wire is fixed, so each particle is located by a single coordinate x_i and its relative kinematics is that of a point on a fixed plane curve. The frame rotates uniformly about a fixed axis through O, so the inertia forces reduce to the centrifugal and Coriolis forces (Slide 28), and the centrifugal force has a potential (Slide 30).
- **Hypotheses:**
  1. The frame $S_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$, with O the vertex of the parabola and $\mathbf{k}_0$ vertical upward, is inertial.
  2. The wire is smooth and bilateral (each particle is a bead): its reaction on P_i is perpendicular to the wire, with components in the plane of the wire and along $\mathbf{j}_1$.
  3. ω is kept constant by the motor; the axis bearings exert no torque about Oz.
  4. Electric repulsion is a central internal force: equal and opposite on the two particles, along $\overrightarrow{P_2P_1}$.
- **Fixed (inertial) frame:** $S_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$, basis $\mathcal{B}_0$, origin at the vertex of the parabola, which lies on the axis and is fixed.
- **Moving frame:** $S_1 = Oxyz = \{O; \mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$, basis $\mathcal{B}_1$, attached to the wire, with the wire in the plane Oxz and $\mathbf{k}_1 = \mathbf{k}_0$ along the axis. The two frames share the origin O. We choose $\mathbf{i}_1 = \mathbf{i}_0$ at $t = 0$.

- **Local basis of the wire at P_i:** unit tangent $\mathbf{t}_i$, in-plane unit normal $\mathbf{n}_i$ and $-\mathbf{j}_1$; deduced in the next section.
- **Degrees of freedom:** two, the abscissas x_1 and x_2 in $Oxyz$ (each bead has 3 coordinates and 2 constraints: $y = 0$ and $z = x^2/(2L)$). In part (e) the symmetry constraint $x_2 = -x_1$ leaves one.

Symbol	Meaning	SI unit
m	mass of each particle	kg
L	parameter of the parabola $z = x^2/(2L)$ (the radius of curvature at the vertex)	m
ω	constant angular velocity of the wire	rad/s
g	gravitational acceleration	m/s^2
k	electric repulsion constant, $F_E = k/d^2$	N·m^2
$\gamma = g/(\omega^2 L)$	weight versus centrifugal force	1
$\kappa = k/(m\omega^2 L^3)$	repulsion versus centrifugal force	1

Change of Basis and Position Vector

The basis $\mathcal{B}_1$ seen from above

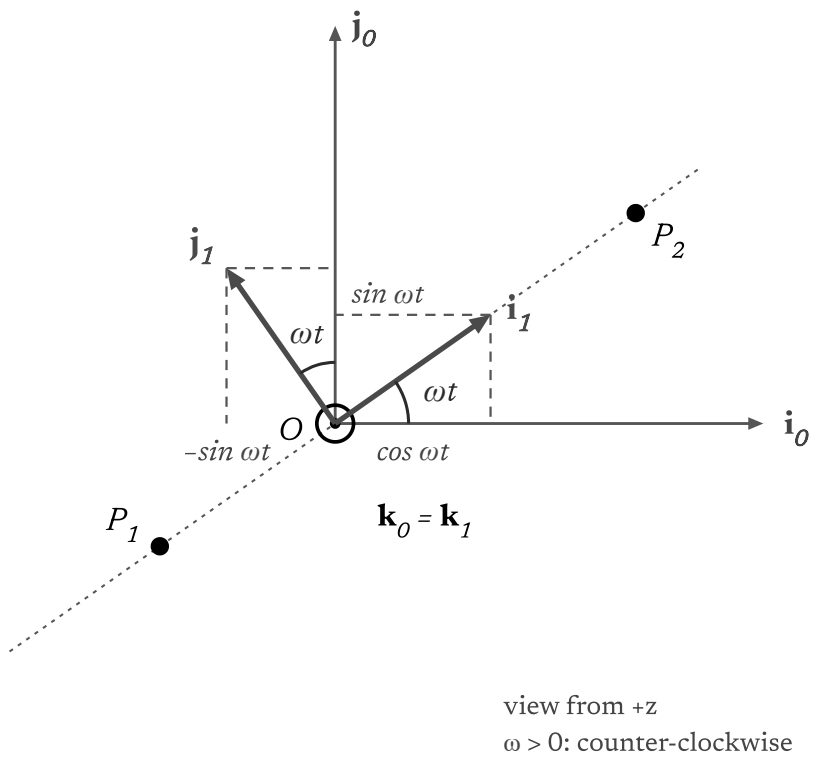


Figure 1. View from $+z$, with $\mathbf{k}_0 = \mathbf{k}_1$ out of the paper ($\odot$) at the vertex O: then $\mathbf{j}_0 = \mathbf{k}_0 \times \mathbf{i}_0$ points up the page. $\mathbf{i}_1, \mathbf{j}_1$ are $\mathbf{i}_0, \mathbf{j}_0$ turned counter-clockwise by ωt; the dashed legs give $\mathbf{i}_1 = \cos\omega t\, \mathbf{i}_0 + \sin\omega t\, \mathbf{j}_0$ and $\mathbf{j}_1 = -\sin\omega t\, \mathbf{i}_0 + \cos\omega t\, \mathbf{j}_0$. The plane of the wire is seen edge-on, along $\mathbf{i}_1$; P_1 ($x_1 < 0$) and P_2 ($x_2 > 0$) project on it.

The rotation is about the vertical $\mathbf{k}_0$, so it is drawn looking down the axis, from $+z$. In that view $\mathbf{k}_0$ points out of the paper, towards the reader ($\odot$). Take $\mathbf{i}_0$ to the right; then $\mathbf{j}_0 = \mathbf{k}_0 \times \mathbf{i}_0$ (right-hand rule: thumb along $\mathbf{k}_0$, towards the reader, fingers turning $\mathbf{i}_0$ counter-clockwise by $90°$) points up the page. With $\boldsymbol{\omega}_{10} = \omega \mathbf{k}_0, \omega > 0$, the wire turns counter-clockwise in this view, and after a time t it has turned the angle ωt:

- $\mathbf{i}_1$ is the hypotenuse (length 1) of a dashed right triangle: its leg adjacent to ωt lies along $\mathbf{i}_0$ and measures $\cos\omega t$; its opposite leg lies along $+\mathbf{j}_0$ and measures $\sin\omega t$. So $\mathbf{i}_1 = \cos\omega t\, \mathbf{i}_0 + \sin\omega t\, \mathbf{j}_0$.
- $\mathbf{j}_1$ starts on $\mathbf{j}_0$ (up the page) and turns counter-clockwise, so its tip moves to the left: its leg along $\mathbf{j}_0$ (adjacent) is $\cos\omega t$ and its leg along $-\mathbf{i}_0$ (opposite) is $\sin\omega t$. So $\mathbf{j}_1 = -\sin\omega t\, \mathbf{i}_0 + \cos\omega t\, \mathbf{j}_0$; **the minus sign comes from the counter-clockwise turn**

carrying the upward axis to the left.

- $\mathbf{k}_1 = \mathbf{k}_0$, the axis of the rotation.

The plane of the wire contains $\mathbf{i}_1$ and $\mathbf{k}_1$, so in this view it is seen edge-on as the dotted line along $\mathbf{i}_1$; P_1 ($x_1 < 0$) and P_2 ($x_2 > 0$) lie on it, on either side of the axis. The columns of $[{}_0R_1]$ are the components of $\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1$ in $\mathcal{B}_0$:

$$[{}_0R_1] = \begin{pmatrix} \cos\omega t & -\sin\omega t & 0 \\ \sin\omega t & \cos\omega t & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = 1\cdot(\cos\omega t\cos\omega t - (-\sin\omega t)\sin\omega t) = \cos^2\omega t + \sin^2\omega t = 1$$

The columns have unit length ($\cos^2\omega t + \sin^2\omega t = 1$) and are mutually orthogonal (first and second: $-\cos\omega t\sin\omega t + \sin\omega t\cos\omega t = 0$; the third is orthogonal to both because their third components are zero), so $[{}_0R_1][{}_0R_1]^T = I$.

- **Motion of S_1: $\mathbf{r}_0^O = \mathbf{v}_0^O = \mathbf{a}_0^O = \mathbf{0}$; $\boldsymbol{\omega}_{10} = \omega\mathbf{k}_1$; $\boldsymbol{\alpha}_{10} = \mathbf{0}$** because ω is constant. Poisson check with the chain rule: $\left.\frac{d\mathbf{i}_1}{dt}\right|_0 = -\omega\sin\omega t\,\mathbf{i}_0 + \omega\cos\omega t\,\mathbf{j}_0 = \omega\,\mathbf{j}_1$, and indeed $\omega\mathbf{k}_1 \times \mathbf{i}_1 = \omega\,\mathbf{j}_1$.

The position vectors and the local basis of the wire

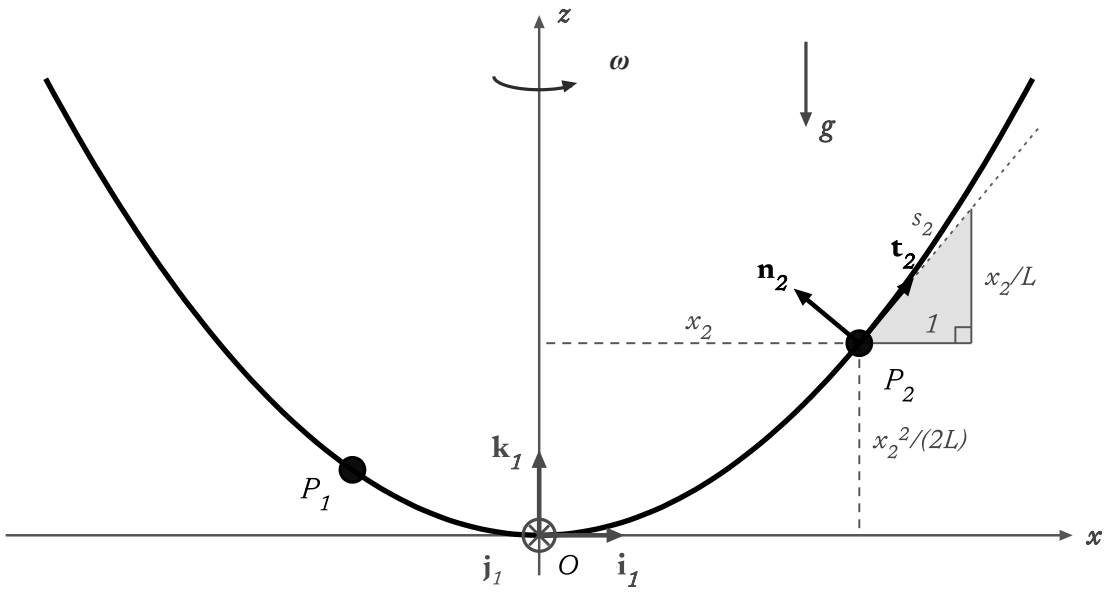


Figure 2. Plane Oxz, fixed in S_1 ($\mathbf{j}_1$ points into the paper, $\otimes$). The position of P_2 has the dashed components x_2 and $z_2 = x_2^2/(2L)$. The shaded slope triangle at P_2 has a horizontal leg proportional to 1, a vertical leg proportional to $dz/dx = x_2/L$ and the hypotenuse $s_2 = \sqrt{1 + x_2^2/L^2}$ along the tangent; dividing by s_2 gives $\mathbf{t}_2$, and turning it 90° towards the concave side gives $\mathbf{n}_2$ (orange).

The plane Oxz is fixed in S_1; it is drawn with $\mathbf{i}_1$ to the right and $\mathbf{k}_1$ up. The direction out of the paper is $\mathbf{i}_1 \times \mathbf{k}_1 = -\mathbf{j}_1$, so $\mathbf{j}_1$ points **into** the paper ($\otimes$).

Position. A bead at abscissa x_i is reached from O by moving x_i along $\mathbf{i}_1$ and then the height of the wire at that abscissa, $z_i = x_i^2/(2L)$, along $\mathbf{k}_1$ (the two dashed components drawn for P_2):

$$\mathbf{r}_1^{P_i} = x_i\,\mathbf{i}_1 + \frac{x_i^2}{2L}\,\mathbf{k}_1, \qquad i = 1, 2$$

Tangent. A tangent vector is the derivative of the position with respect to the coordinate; with $\frac{d}{dx}\left(\frac{x^2}{2L}\right) = \frac{2x}{2L} = \frac{x}{L}$:

$$\frac{d}{dx}\left(x\,\mathbf{i}_1 + \frac{x^2}{2L}\mathbf{k}_1\right) = \mathbf{i}_1 + \frac{x}{L}\,\mathbf{k}_1$$

It is the hypotenuse of the shaded slope triangle at P_2: horizontal leg 1, vertical leg $dz/dx = x_i/L$, hence length $s_i = \sqrt{1 + x_i^2/L^2}$ (Pythagoras). Dividing by s_i:

$$\mathbf{t}_i = \frac{1}{s_i}\left(\mathbf{i}_1 + \frac{x_i}{L}\mathbf{k}_1\right), \qquad \mathbf{n}_i = \frac{1}{s_i}\left(-\frac{x_i}{L}\mathbf{i}_1 + \mathbf{k}_1\right)$$

$\mathbf{n}_i$ is $\mathbf{t}_i$ turned 90° counter-clockwise in the drawing (the components (a, b) become $(-b, a)$), which points towards the concave side of the parabola, upwards and towards the axis. They are orthogonal, $\mathbf{t}_i \cdot \mathbf{n}_i = \frac{1}{s_i^2}\left(-\frac{x_i}{L} + \frac{x_i}{L}\right) = 0$, both have length $\sqrt{1 + x_i^2/L^2}/s_i = 1$, and, using $\mathbf{i}_1 \times \mathbf{k}_1 = -\mathbf{j}_1$ and $\mathbf{k}_1 \times \mathbf{i}_1 = \mathbf{j}_1$,

$$\mathbf{t}_i \times \mathbf{n}_i = \frac{1}{s_i^2}\left[\mathbf{i}_1 \times \mathbf{k}_1 - \frac{x_i^2}{L^2}\,\mathbf{k}_1 \times \mathbf{i}_1\right] = \frac{1}{s_i^2}\left[-\mathbf{j}_1 - \frac{x_i^2}{L^2}\mathbf{j}_1\right] = -\mathbf{j}_1$$

(the direction out of the paper), so $\{\mathbf{t}_i, \mathbf{n}_i, -\mathbf{j}_1\}$ is a right-handed orthonormal triad attached to the wire at P_i. The reaction of the smooth wire lies in the plane spanned by $\mathbf{n}_i$ and $\mathbf{j}_1$.

The chord between the beads

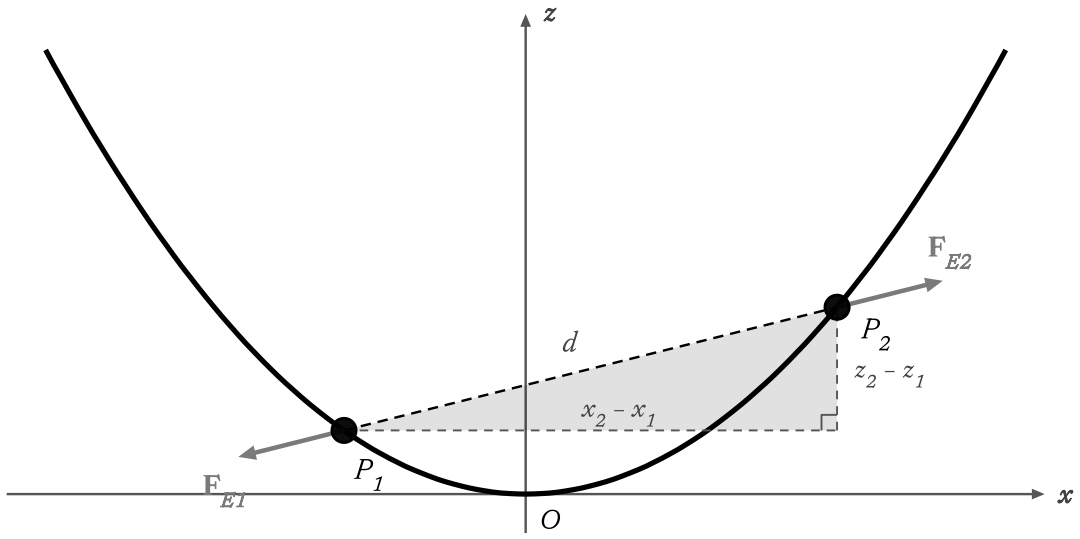


Figure 3. Chord P_1P_2: in the shaded right triangle the horizontal leg is $x_2 - x_1$, the vertical leg is $z_2 - z_1 = (x_2^2 - x_1^2)/(2L) = (x_2 - x_1)(x_1 + x_2)/(2L)$ and the hypotenuse is the distance d; the slope of the chord is $(x_1 + x_2)/(2L)$. The repulsions $\mathbf{F}_{E1}$ and $\mathbf{F}_{E2} = -\mathbf{F}_{E1}$ (pink) act along the chord, pushing the beads apart.

The repulsion acts along the chord P_1P_2. With $\Delta x = x_1 - x_2$, the shaded right triangle has the horizontal leg $|\Delta x|$, the vertical leg $\Delta z = z_1 - z_2 = \dfrac{x_1^2 - x_2^2}{2L} = \Delta x\,\dfrac{x_1 + x_2}{2L}$ (difference of squares) and the hypotenuse d. Hence

$$\overrightarrow{P_2P_1} = \mathbf{r}_1^{P_1} - \mathbf{r}_1^{P_2} = \Delta x\left[\mathbf{i}_1 + \frac{x_1 + x_2}{2L}\mathbf{k}_1\right], \qquad d^2 = |\overrightarrow{P_2P_1}|^2 = \Delta x^2\left[1 + \frac{(x_1 + x_2)^2}{4L^2}\right]$$

so the slope of the chord is $(x_1 + x_2)/(2L)$, the mean of the slopes x_i/L at its ends.

Kinematics

(a) Relative position, velocity and acceleration

Pedagogical Justification: Relative kinematics is computed in S_1, where $\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1$ are constant, so only the scalar components are differentiated. Each $z_i = x_i^2/(2L)$ depends on time through $x_i(t)$, so the chain rule is needed.

Chain rule: $\dfrac{d}{dt}\left(\dfrac{x_i^2}{2L}\right) = \dfrac{2x_i}{2L}\,\dot{x}_i = \dfrac{x_i \dot{x}_i}{L}$. Hence

$$\mathbf{v}_1^{P_i} = \dot{x}_i\,\mathbf{i}_1 + \frac{x_i \dot{x}_i}{L}\,\mathbf{k}_1$$

Product rule on $x_i \dot{x}_i$: $\dfrac{d}{dt}(x_i \dot{x}_i) = \dot{x}_i \dot{x}_i + x_i \ddot{x}_i$. Hence

$$\mathbf{a}_1^{P_i} = \ddot{x}_i\,\mathbf{i}_1 + \frac{\dot{x}_i^2 + x_i \ddot{x}_i}{L}\,\mathbf{k}_1$$

Note $\mathbf{v}_1^{P_i} = s_i \dot{x}_i\,\mathbf{t}_i$, tangent to the wire, as it must be.

Forces and Equations of Motion

(b) Newton's second law in $Oxyz$

Pedagogical Justification: $Oxyz$ is not inertial; Newton's law holds there if the inertia forces are added, $m\mathbf{a}_1^{P_i} = \mathbf{F} + \mathbf{F}_i$ (Notes Eqs. 4.62–4.63). Projecting on the tangent $\mathbf{t}_i$ eliminates the unknown reaction and gives the equation of motion; the other two projections give the reaction.

Actual forces on P_i:

- Weight: $-mg\,\mathbf{k}_1$.
- Reaction of the smooth wire: $\mathbf{N}_i = N_{in}\,\mathbf{n}_i + N_{iy}\,\mathbf{j}_1$ (so $\mathbf{N}_i \cdot \mathbf{t}_i = 0$).
- Electric repulsion. With $\overrightarrow{P_2P_1}$ and d of the chord triangle, the force on P_1 has magnitude k/d^2 and the direction of $\overrightarrow{P_2P_1}/d$, and the force on P_2 is opposite:

$$\mathbf{F}_{E1} = \frac{k}{d^2}\,\frac{\overrightarrow{P_2P_1}}{d} = \frac{k}{d^3}\,\Delta x\left[\mathbf{i}_1 + \frac{x_1+x_2}{2L}\mathbf{k}_1\right] = -\mathbf{F}_{E2}$$

Inertia forces ($\mathbf{a}_0^O = \mathbf{0}$, $\boldsymbol{\alpha}_{10} = \mathbf{0}$, so $\mathbf{F}_A = \mathbf{F}_E^{\text{Euler}} = \mathbf{0}$):

- Centrifugal: $\boldsymbol{\omega}_{10} \times \mathbf{r}_1^{P_i} = \omega\mathbf{k}_1 \times \left(x_i\mathbf{i}_1 + \frac{x_i^2}{2L}\mathbf{k}_1\right) = \omega x_i\,\mathbf{j}_1$, then $\boldsymbol{\omega}_{10} \times \omega x_i\mathbf{j}_1 = \omega^2 x_i\,(\mathbf{k}_1 \times \mathbf{j}_1) = -\omega^2 x_i\,\mathbf{i}_1$, so $\mathbf{F}_{C,i} = m\omega^2 x_i\,\mathbf{i}_1$ (outward from the axis, Slide 30).
- Coriolis: $\boldsymbol{\omega}_{10} \times \mathbf{v}_1^{P_i} = \omega\mathbf{k}_1 \times \left(\dot{x}_i\mathbf{i}_1 + \frac{x_i\dot{x}_i}{L}\mathbf{k}_1\right) = \omega\dot{x}_i\,\mathbf{j}_1$, so $\mathbf{F}_{COR,i} = -2m\omega\dot{x}_i\,\mathbf{j}_1$.

(b, continued) Newton's equations

Newton's second law for each particle (vector form):

$$m\left[\ddot{x}_i\,\mathbf{i}_1 + \frac{\dot{x}_i^2 + x_i\ddot{x}_i}{L}\,\mathbf{k}_1\right] = -mg\,\mathbf{k}_1 + N_{in}\,\mathbf{n}_i + N_{iy}\,\mathbf{j}_1 + \mathbf{F}_{Ei} + m\omega^2 x_i\,\mathbf{i}_1 - 2m\omega\dot{x}_i\,\mathbf{j}_1, \qquad i = 1, 2$$

Projection on $s_i\mathbf{t}_i = \mathbf{i}_1 + \frac{x_i}{L}\mathbf{k}_1$ (no reaction). Left side: $m\left[\ddot{x}_i + \frac{x_i}{L}\cdot\frac{\dot{x}_i^2 + x_i\ddot{x}_i}{L}\right] = m\left[\left(1 + \frac{x_i^2}{L^2}\right)\ddot{x}_i + \frac{x_i\dot{x}_i^2}{L^2}\right]$. Right side: weight $-mg\,x_i/L$; centrifugal $m\omega^2 x_i$; Coriolis 0; electric, for P_1,

$$\mathbf{F}_{E1}\cdot\left(\mathbf{i}_1 + \frac{x_1}{L}\mathbf{k}_1\right) = \frac{k}{d^3}\,(x_1 - x_2)\left[1 + \frac{x_1(x_1+x_2)}{2L^2}\right]$$

and for P_2 the same with the indices exchanged (because $\mathbf{F}_{E2} = -\mathbf{F}_{E1}$ and $-\Delta x = x_2 - x_1$). The **equations of motion** are

$$\boxed{m\left[\left(1 + \frac{x_1^2}{L^2}\right)\ddot{x}_1 + \frac{x_1\dot{x}_1^2}{L^2}\right] = m\left(\omega^2 - \frac{g}{L}\right)x_1 + \frac{k}{d^3}\,(x_1 - x_2)\left[1 + \frac{x_1(x_1+x_2)}{2L^2}\right]}$$

$$\boxed{m\left[\left(1+\frac{x_2^2}{L^2}\right)\ddot{x}_2+\frac{x_2\dot{x}_2^2}{L^2}\right]=m\left(\omega^2-\frac{g}{L}\right)x_2+\frac{k}{d^3}\left(x_2-x_1\right)\left[1+\frac{x_2(x_1+x_2)}{2L^2}\right]}$$

with $d^3=|x_1-x_2|^3\left[1+\dfrac{(x_1+x_2)^2}{4L^2}\right]^{3/2}$.

Projection on $\mathbf{j}_1$: $0=N_{iy}-2m\omega\dot{x}_i$, so $\boxed{N_{iy}=2m\omega\dot{x}_i}$: the wire pushes each bead sideways to cancel the Coriolis force.

Projection on $s_i\mathbf{n}_i=-\dfrac{x_i}{L}\mathbf{i}_1+\mathbf{k}_1$. Left side: $m\left[-\dfrac{x_i\ddot{x}_i}{L}+\dfrac{\dot{x}_i^2+x_i\ddot{x}_i}{L}\right]=\dfrac{m\dot{x}_i^2}{L}$. Right side: $N_{in}s_i-mg-\dfrac{m\omega^2x_i^2}{L}+\mathbf{F}_{Ei}\cdot(s_i\mathbf{n}_i)$, where for P_1

$$\mathbf{F}_{E1}\cdot\left(-\frac{x_1}{L}\mathbf{i}_1+\mathbf{k}_1\right)=\frac{k\,\Delta x}{d^3}\left[-\frac{x_1}{L}+\frac{x_1+x_2}{2L}\right]=\frac{k\,\Delta x}{d^3}\cdot\frac{x_2-x_1}{2L}=-\frac{k\,\Delta x^2}{2Ld^3}$$

and the same value is found for P_2. Therefore

$$\boxed{N_{in}=\frac{m}{s_i}\left[\frac{\dot{x}_i^2}{L}+g+\frac{\omega^2x_i^2}{L}\right]+\frac{k\,(x_1-x_2)^2}{2L\,s_i\,d^3}},\qquad s_i=\sqrt{1+\frac{x_i^2}{L^2}}$$

Answers and Checks

(a) Relative position, velocity and acceleration

$$\mathbf{r}_1^{P_i}=x_i\,\mathbf{i}_1+\frac{x_i^2}{2L}\,\mathbf{k}_1,\qquad\mathbf{v}_1^{P_i}=\dot{x}_i\,\mathbf{i}_1+\frac{x_i\dot{x}_i}{L}\,\mathbf{k}_1,\qquad\mathbf{a}_1^{P_i}=\ddot{x}_i\,\mathbf{i}_1+\frac{\dot{x}_i^2+x_i\ddot{x}_i}{L}\,\mathbf{k}_1\qquad(i=1,2)$$

(b) Newton's second law in $Oxyz$

The vector equation of each bead, with its tangential projections (the two boxed equations of motion), $N_{iy} = 2m\omega\dot{x}_i$ and the boxed N_{in}, all derived in the previous section.

(c) Motor torque that keeps ω constant

> **Pedagogical Justification:** The motor torque acts on the wire; the wire acts on the particles through the reactions $\mathbf{N}_i$, and the particles act back with $-\mathbf{N}_i$. The wire is a massless guide (if it had mass and were symmetric about Oz, rotating at constant ω, its own angular momentum about Oz would be constant; this remark goes beyond the particle units and does not change the result). Hence the moment about Oz of all the actions on it must vanish.

Balance of moments about Oz on the wire: $M + \sum_i \left[\mathbf{r}^{P_i} \times (-\mathbf{N}_i)\right] \cdot \mathbf{k}_1 = 0$. Expand the cross product with $\mathbf{r}^{P_i} = x_i\mathbf{i}_1 + z_i\mathbf{k}_1$ and $\mathbf{N}_i = N_{ix}\mathbf{i}_1 + N_{iy}\mathbf{j}_1 + N_{iz}\mathbf{k}_1$; its $\mathbf{k}_1$ component is $x_i N_{iy} - 0 \cdot N_{ix} = x_i N_{iy}$. With $N_{iy} = 2m\omega\dot{x}_i$ from part (b):

$$\boxed{M = x_1 N_{1y} + x_2 N_{2y} = 2m\omega\left(x_1\dot{x}_1 + x_2\dot{x}_2\right)}$$

Check with the angular momentum of the particles in S_0. By Notes Eq. 3.26, $\mathbf{v}_0^{P_i} = \mathbf{v}_1^{P_i} + \boldsymbol{\omega}_{10} \times \mathbf{r}_1^{P_i} = \dot{x}_i\mathbf{i}_1 + \omega x_i\mathbf{j}_1 + \frac{x_i\dot{x}_i}{L}\mathbf{k}_1$, whose $\mathbf{k}_1$-moment about O is $m\left[x_i(\omega x_i) - 0\cdot\dot{x}_i\right] = m\omega x_i^2$. So $H_{Oz} = m\omega(x_1^2 + x_2^2)$ and, by the product and chain rules, $\dot{H}_{Oz} = 2m\omega(x_1\dot{x}_1 + x_2\dot{x}_2)$. The weights are parallel to Oz and the electric pair is central ($(\mathbf{r}^{P_1} - \mathbf{r}^{P_2}) \times \mathbf{F}_{E1} = \mathbf{0}$), so only the reactions have a moment about Oz, and that moment equals M. Both routes agree with the official key (Problems.pdf, p. 75).

(d) Equilibrium configurations in $Oxyz$

> **Pedagogical Justification:** A relative equilibrium has $\dot{x}_i = \ddot{x}_i = 0$; then the two equations of motion become algebraic. Writing them with $u_i = x_i/L$ and dividing by $m\omega^2 L$ shows that only γ and κ matter.

With $\dot{x}_i = \ddot{x}_i = 0$, $x_i = Lu_i$ and $d = LD$, $D = |u_1 - u_2|\sqrt{1 + (u_1+u_2)^2/4}$, dividing by $m\omega^2 L$:

$$f_1 \equiv (1-\gamma)\,u_1 + \frac{\kappa}{D^3}(u_1 - u_2)\left[1 + \frac{u_1(u_1+u_2)}{2}\right] = 0, \qquad f_2 \equiv (1-\gamma)\,u_2 + \frac{\kappa}{D^3}(u_2 - u_1)\left[1 + \frac{u_2(u_1+u_2)}{2}\right] = 0$$

Sum. With $S = u_1 + u_2$, the electric terms add to $\dfrac{\kappa}{D^3}(u_1 - u_2)\left[\dfrac{u_1 S}{2} - \dfrac{u_2 S}{2}\right] = \dfrac{\kappa\,(u_1 - u_2)^2 S}{2D^3}$, so

$$f_1 + f_2 = S\left[(1-\gamma) + \frac{\kappa\,(u_1-u_2)^2}{2D^3}\right] = 0$$

Either $S = 0$ (symmetric configuration) or $\gamma - 1 = \dfrac{\kappa (u_1-u_2)^2}{2D^3} > 0$.

Case $\gamma < 1$. If $S = 0$, $u_2 = -u_1$, $D = 2|u_1|$ and $f_1 = u_1\left[(1-\gamma) + \dfrac{2\kappa}{8|u_1|^3}\right]$: both terms of the bracket are positive and $u_1 \neq 0$ (the particles cannot coincide), so $f_1 \neq 0$. If $S \neq 0$ the bracket of the sum is positive. **No equilibrium exists.** (Centrifugal force beats gravity everywhere and the repulsion only pushes the particles further out.)

Case $\gamma = 1$. Both alternatives require $\kappa = 0$. For $\kappa > 0$ there is no equilibrium; only without repulsion ($\kappa = 0$) is every configuration an equilibrium (gravity and centrifugal force balance along the wire at every point: $(\omega^2 - g/L)x_i = 0$).

Case $\gamma > 1$, symmetric ($x_2 = -x_1$). $f_1 = u_1\left[(1-\gamma) + \dfrac{\kappa}{4|u_1|^3}\right] = 0$ gives $|u_1|^3 = \dfrac{\kappa}{4(\gamma-1)}$:

$$\boxed{x_1 = -x_2, \qquad x_1^2 = \left[\frac{\kappa L^3}{4(\gamma-1)}\right]^{2/3}}$$

Case $\gamma > 1$, asymmetric ($S \neq 0$). Insert $\dfrac{\kappa}{D^3} = \dfrac{2(\gamma-1)}{(u_1-u_2)^2}$ from the sum into $f_1 = 0$ and divide by $(\gamma - 1)$:

$$-u_1 + \frac{2}{u_1 - u_2}\left[1 + \frac{u_1(u_1+u_2)}{2}\right] = 0 \implies u_1(u_1-u_2) = 2 + u_1(u_1+u_2) \implies \boxed{x_1 x_2 = -L^2}$$

The particles are on opposite branches and the chord P_1P_2 passes through the focus $(0, L/2)$ of the parabola. Then $u_2 = -1/u_1$, $S = u_1 - 1/u_1$, $u_1 - u_2 = u_1 + 1/u_1$, and $1 + S^2/4 = (u_1 + 1/u_1)^2/4$, so $D^3 = |u_1 + 1/u_1|^6/8$ and the sum condition becomes

$$\gamma - 1 = \frac{4\kappa}{(u_1 + 1/u_1)^4} \implies \left| u_1 + \frac{1}{u_1} \right| = \left(\frac{4\kappa}{\gamma - 1} \right)^{1/4}$$

Since $|u_1 + 1/u_1| \geq 2$, with equality only for $|u_1| = 1$ (which is the symmetric equilibrium), **asymmetric equilibria exist if and only if** $\kappa > 4(\gamma - 1)$, i.e. exactly when the symmetric equilibrium lies at $|x_1| > L$. They come in mirror pairs, $u_1 = \frac{1}{2}\left[q \pm \sqrt{q^2 - 4}\right]$ with $q = (4\kappa/(\gamma-1))^{1/4}$ and $u_2 = -1/u_1$ (plus the configurations with the labels exchanged). No equilibrium has both particles on the same branch.

(e) Symmetric motion with $\gamma = \kappa = 2$: turning points

> **Pedagogical Justification:** In $Oxyz$ the wire (and the symmetry constraint) is fixed, $\boldsymbol{\omega}_{10}$ is constant and the origin is fixed. The reactions and the Coriolis forces do no work in S_1 (Slide 31), and the weights, the electric pair and the centrifugal forces have potentials (Slides 29–30). The mechanical energy of the pair in S_1 is therefore conserved, and its turning points ($\dot{x}_1 = 0$) give the extreme values of x_1.

Put $x_1 = x > 0$, $x_2 = -x$ (the repulsion becomes infinite at $x = 0$, so x_1 stays positive). Then $d = 2x$ and, for each particle, $|\mathbf{v}_1^{P_i}|^2 = \dot{x}^2\left(1 + x^2/L^2\right)$.

- Kinetic energy: $T_1 = 2 \cdot \frac{1}{2} m \dot{x}^2 \left(1 + \dfrac{x^2}{L^2}\right) = m\dot{x}^2 \left(1 + \dfrac{x^2}{L^2}\right)$.
- Gravity: $2\,mg\,\dfrac{x^2}{2L} = \dfrac{mgx^2}{L}$. Centrifugal: $2\left(-\frac{1}{2}m\omega^2 x^2\right) = -m\omega^2 x^2$. Electric: $\dfrac{k}{d} = \dfrac{k}{2x}$ (its gradient gives the repulsion k/d^2).

$$E_1 = m\dot{x}^2\left(1 + \frac{x^2}{L^2}\right) + m\left(\frac{g}{L} - \omega^2\right)x^2 + \frac{k}{2x} = \text{const}$$

Initial value. $|\mathbf{v}_1^{P_1}(0)| = L\omega$ at $x = L$ gives $2\dot{x}(0)^2 = L^2\omega^2$, so $T_1(0) = mL^2\omega^2$ and $E_1 = mL^2\omega^2 + m(g/L - \omega^2)L^2 + k/(2L)$. Dividing by $m\omega^2 L^2$, with $u = x/L$:

$$\frac{E_1}{m\omega^2 L^2} = \gamma + \frac{\kappa}{2} = 3 \qquad (\gamma = \kappa = 2)$$

Turning points ($\dot{x} = 0$): $(\gamma - 1)u^2 + \dfrac{\kappa}{2u} = \gamma + \dfrac{\kappa}{2}$, i.e. $u^2 + \dfrac{1}{u} = 3$. Multiply by $u > 0$:

$$\boxed{\frac{x_1^3}{L^3} - 3\,\frac{x_1}{L} + 1 = 0}$$

Roots. With $u = 2\cos\theta$ and $4\cos^3\theta - 3\cos\theta = \cos 3\theta$: $8\cos^3\theta - 6\cos\theta + 1 = 2\cos 3\theta + 1 = 0$, so $\cos 3\theta = -\tfrac{1}{2}$ and $\theta = 2\pi/9,\ 4\pi/9,\ 8\pi/9$. The roots are $u = 2\cos(2\pi/9) = 1.532$, $2\cos(4\pi/9) = 0.347$ and $2\cos(8\pi/9) = -1.879$ (rejected, $x_1 > 0$). Since the starting point $u = 1$ lies between the two positive roots (there $u^2 + 1/u = 2 < 3$),

$$x_{1,\min} = 2L\cos\frac{4\pi}{9} \approx 0.347\,L, \qquad x_{1,\max} = 2L\cos\frac{2\pi}{9} \approx 1.532\,L$$

and x_1 oscillates between them. The key also lists $x_1^3/L^3 - 3x_1/L - 1 = 0$ for $x_1 < 0$: that is the mirror image (a start on the negative branch) and does not apply to $x_1(0) = L > 0$.

Checks

- **Effective spring along the wire.** Gravity and the centrifugal force combine into $m(\omega^2 - g/L)x_i$ in the tangential equation: a restoring force if $\gamma > 1$ (gravity wins and pulls the beads to the vertex) and a repelling one if $\gamma < 1$. Equilibrium requires the repulsion to be balanced by this restoring force, hence the condition $\gamma > 1$.
- **Limits.** $\kappa \to 0$: the symmetric equilibrium collapses to $x_1 = x_2 = 0$ (no repulsion to separate the beads). $\gamma \to 1^+$: $x_1^2 \to \infty$, the restoring force vanishes and the beads are pushed to infinity. At the vertex, $x_i = 0$ gives $\mathbf{t}_i = \mathbf{i}_1$ and $\mathbf{n}_i = \mathbf{k}_1$, as the slope triangle degenerates to its horizontal leg.
- **Torque sign.** $M = \frac{d}{dt}\left[m\omega(x_1^2 + x_2^2)\right]$: when the beads move outward the moment of inertia of the system grows and the motor has to supply angular momentum ($M > 0$); when they move inward it must brake ($M < 0$).
- **Orthogonality:** $\mathbf{t}_i \cdot \mathbf{n}_i = 0$, $\mathbf{t}_i \times \mathbf{n}_i = -\mathbf{j}_1$ and $\det[{}_0R_1] = 1$ (change-of-basis section).
- **Inertial check.** Differentiating $[{}_0R_1]\mathbf{r}_1^{P_i}$ twice reproduces $\mathbf{a}_1^{P_i} + \boldsymbol{\omega}_{10} \times (\boldsymbol{\omega}_{10} \times \mathbf{r}_1^{P_i}) + 2\boldsymbol{\omega}_{10} \times \mathbf{v}_1^{P_i}$ (Notes Eq. 3.27), and $dE_1/dt = 0$ along the equations of motion of part (b) (both checked symbolically with sympy, together with the equilibria of part (d)).
- **Dimensions.** $[x_i^2/(2L)] = \mathrm{m}$, $[x_i/L] = 1$; $[k/d^2] = \mathrm{N \cdot m^2/m^2} = \mathrm{N}$; $[m\omega^2 x] = \mathrm{kg \cdot s^{-2} \cdot m} = \mathrm{N}$; $[M] = [m\omega x\dot{x}] = \mathrm{kg \cdot s^{-1} \cdot m \cdot m/s} = \mathrm{N \cdot m}$; $[\gamma] = \dfrac{\mathrm{m/s^2}}{\mathrm{s^{-2}\,m}} = 1$; $[\kappa] = \dfrac{\mathrm{N \cdot m^2}}{\mathrm{kg \cdot s^{-2} \cdot m^3}} = 1$.

- **Check against the official key:** the key of `Problems.pdf` gives the motor torque $2m\omega\left(\dot{x}_1 x_1 + \dot{x}_2 x_2\right)$, the equilibrium cases ($\gamma < 1$: none; $\gamma = 1$: only for $\kappa = 0$; $\gamma > 1$: symmetric with $x_1 = -x_2$ and $x_1^2 = \left[\kappa L^3 / \left(4(\gamma - 1)\right)\right]^{2/3}$, and asymmetric ones) and the turning-point equations $x_1^3/L^3 - 3x_1/L \pm 1 = 0$. They coincide with the results of (c), (d) and (e) (for $\gamma = \kappa = 2$ the turning-point condition $u^3 - 3u + 1 = 0$ is the positive-x_1 case).

Source: Problems.pdf (PDF page 21); key on PDF page 75